

2 Add the arms

Next replace the code you originally had between the comment line **# arms** and the comment line **# neck** with the code shown here. It uses the arm function to draw three arms.

```
# arms
```

```
t.goto(-90, 85)
```

```
t.setheading(180)
```

```
arm('light blue')
```

```
t.goto(-90, 20)
```

```
t.setheading(180)
```

```
arm('purple')
```

```
t.goto(10, 85)
```

```
t.setheading(0)
```

```
arm('goldenrod')
```

Set the turtle to point to the robot's right (the left edge of the window).

Use the arm function to draw a light blue arm.

Set the turtle to point to the robot's left (the right edge of the window).

▽ Moving arms

Now that you can draw a whole arm in one go, you can change its position so the robot looks like it's scratching its head or maybe dancing a Highland Fling! To do this, use the **setheading()** function to change the direction the turtle is facing when it starts to draw the arm.

```
# arms
```

```
t.goto(-90, 80)
```

```
t.setheading(135)
```

```
arm('hot pink')
```

```
t.goto(10, 80)
```

```
t.setheading(315)
```

```
arm('hot pink')
```

Set the turtle to point to the top-left corner of the window.

Use the arm function to draw an arm on the right.

Set the turtle to point to the bottom-right corner of the window.

Use the arm function to draw an arm on the left.



EXPERT TIPS

Trial and error

When you're designing a robot or adding new features to an existing robot, it may take a bit of trial and error to get things just how you want them. If you add the lines **print(t.window_width())** and **print(t.window_height())** after the line **t.speed('slowest')**, Python will display the height and width of your Turtle window in the shell. Then mark out a grid of that size on graph-paper to help you to work out the coordinates of each body part.

